GROUP PROJECT CHECKPOINT

Predicting 30-Day Customer Repeat Purchases Using Behavioural and Semantic Features

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GROUP PROGRESS AND DATA AUDIT

Project question: Can a customer's merchandise transactions in the previous 90 days predict whether they will make another merchandise purchase in the next 30 days? The project uses UCI Online Retail (ID 352): 541,909 invoice lines dated 1 December 2010 to 9 December 2011. The raw audit found 135,080 missing customer IDs, 1,454 missing descriptions and 5,268 exact duplicate rows.

DATA CLEANING AND MY CURRENT FOCUS

- Exact duplicate invoice lines are removed, retaining one copy so the same transaction is not counted twice. Invoice and stock-code text is standardised; dates and numeric fields are parsed; descriptions are uppercased, punctuation is removed and whitespace is normalised.
- Rows with missing customer IDs will not be considered as a purchase. Because the task predicts repeat purchases for identifiable individuals, a row without an ID cannot be linked reliably to a customer's prior activity or future outcome. These rows are excluded from customer snapshots, model features and labels; they remain in the raw audit and are not treated as non-returning customers.
- An eligible purchase must have an identified customer, valid timestamp, positive quantity and price, and not be a cancellation. Invoice numbers beginning with "C" and negative-quantity lines are not counted as purchases. Identified merchandise returns are retained separately for return auditing, but are excluded from positive-purchase features and labels.
- Postage, carriage, manual adjustments, fees, bank charges and gift vouchers are retained for revenue auditing but excluded from product features and repeat-purchase labels because they are not merchandise purchases. Missing or unusable descriptions are excluded from product-text features.
- The notebook reports 536,641 cleaned rows (without duplicates); 392,692 valid identified positive-purchase lines; 391,153 merchandise lines; 1,539 transaction-charge lines; 8,506 identified merchandise-return lines; and 135,077 other invalid or unidentified lines. The last figure is a post-cleaning category - not simply the raw missing-ID count - and reflects the purchase eligibility rules.

MODELLING DESIGN AND PRELIMINARY FINDINGS

Each row represents an identified customer at a month-end snapshot. A customer is included only after at least one merchandise purchase in the preceding 90-day feature window. The label is whether they purchase merchandise in the following 30 days; target-period transactions do not enter feature calculations. There are 16,123 customer-snapshot rows across eight monthly cohorts. March-June train; July selects the model and threshold; August-October are the rolling tests. Models compare behavioural/cancellation features with TF-IDF/LSA and MiniLM product-description representations. The text representations were checked before their model gains were interpreted: LSA-TF-IDF produces markedly tighter and more balanced clusters than raw sparse KMeans, while MiniLM is weaker on both silhouette and cluster balance, including in a matched-K comparison.

Model or Baseline	July PR-AUC	Aug-Oct PR-AUC
Majority baseline	0.317	0.406
Recency rule	0.407	0.418
Logistic RFM	0.598	0.625
Boosting, behaviour	0.608	0.633
Boosting + LSA-TFIDF	0.615	0.634
Boosting + MiniLM	0.612	0.634

The July selection settled on Boosting + LSA-TF-IDF, which recorded a validation PR-AUC of 0.615, a pooled PR-AUC of 0.634 on the final test, and a lift of 1.77 at the 20 percent mark. Every learned model exceeds both the majority baseline and the recency rule, while adding text representations yields only marginal improvement over behaviour-only boosting. Because the final test plays no part in July selection and the paired intervals for the leading learned models overlap, the report regards them as statistically equivalent rather than declaring a winner on the basis of a third-decimal gap.

INDIVIDUAL CONTRIBUTIONS

Member	Specific work completed / status	Evidence / output	Approx. hours	Next task
Stacia Benita	Assigned Stages 1-3: data audit, cleaning and invoice build. Shared outputs cover missingness, duplicate sensitivity, product/charge taxonomy and revenue audit	EDA figures, cleaning audit and revenue audit.	18 h	Independently rerun Stages 1-3; verify duplicate and charge-line decisions.
Devano Jonathan	Reviewed text-feature methodology; replaced raw sparse TF-IDF clustering with LSA-TF-IDF; checked matched K=16 and MiniLM clusters; reviewed October errors; prepared checkpoint.	Cluster-quality tables, cluster examples, error summary and 19 passing tests.	20 h	Accept cluster descriptions and review final wording.
Surjadjajadi Clara Natalie	Completed shared snapshot, model and uncertainty outputs; final pipeline review and report integration remain.	Validation, monthly results and paired interval files.	19 h	Verify pipeline section and merge final report.

NEXT STEPS

Target	Owner	Task
6-7 Nov 2026	All members	Finish personal evidence checks; submit each individual two-page checkpoint with confirmed hours
9-14 Nov 2026	All members	Reconcile sensitivity and cluster-count results, finalise figures and maintain iteration log.
15-20 Nov 2026	All members	Merge report sections with consistent claims and caveats.
21-25 Nov 2026	All members	Build slides, rehearse and check numerical consistency.

BLOCKERS / SUPPORT NEEDED

Each member must confirm their own approximate hours and personally verify assigned evidence. Cluster labels still need human acceptance.

AI USE AND VERIFICATION

AI assistance was used throughout the project to support data analysis and code generation. I used GPT-Luna 6 to analyse the dataset for duplicated rows and missing values, to review the data-cleaning process for potentially missed issues, and to refine the reasoning for including or excluding data; it was also used to generate Python code for the analyses and visualisations based on the comparison methods chosen by the team. All AI-generated suggestions and code were reviewed, verified, and adjusted appropriately as well as checked against the archived dataset and saved result tables, with the notebook run end to end and all 19 contract tests passing. Interpretation of results and final submission wording remain the responsibility of the team members, who verify their assigned analyses and confirm that the final claims are supported by the evidence.